

Module 3

Ordinary differential equations of first order

3.1 Linear and Bernoulli's differential equations

Introduction:

1. Linear differential equation in y :

This is of the form $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x alone.

General solution is $y \cdot IF = \int Q \cdot IF \, dx + c$, where $IF = e^{\int P \, dx}$.

2. Linear differential equation in x :

This is of the form $\frac{dx}{dy} + Px = Q$, where P and Q are functions of y alone.

General solution is $x \cdot IF = \int Q \cdot IF \, dy + c$, where $IF = e^{\int P \, dy}$.

3. Bernoulli's differential equation in y :

This is of the form $\frac{dy}{dx} + Py = Qy^n$, where P and Q are functions of x alone.

Dividing this equation by y^n , $y^{-n} \frac{dy}{dx} + Py^{1-n} = Q$ ---- (1)

Put $y^{1-n} = t$, then $(1-n)y^{-n} \frac{dy}{dx} = \frac{dt}{dx}$

Equation (1) becomes $\frac{1}{1-n} \frac{dt}{dx} + Pt = Q$

Reduced linear differential equation is $\frac{dt}{dx} + (1-n)Pt = (1-n)Q$

4. Bernoulli's differential equation in x :

This is of the form $\frac{dx}{dy} + Px = Qx^n$, where P and Q are functions of y alone.

Dividing this equation by x^n , $x^{-n} \frac{dx}{dy} + Px^{1-n} = Q$ ---- (1)

Put $x^{1-n} = t$, then $(1-n)x^{-n} \frac{dx}{dy} = \frac{dt}{dy}$

Equation (1) becomes $\frac{1}{1-n} \frac{dt}{dy} + Pt = Q$

Reduced linear differential equation is $\frac{dt}{dy} + (1-n)Pt = (1-n)Q$

Problems:

1. Solve $\frac{dy}{dx} + y \cot x = \cos x$

This is a linear L.D.E in y with $P = \cot x, Q = \cos x$

$$IF = e^{\int P dx} = e^{\int \cot x dx} = \sin x$$

General solution is $y \cdot IF = \int Q \cdot IF dx + c$

$$y \cdot \sin x = \int \cos x \cdot \sin x dx + c$$

$$y \sin x = \frac{1}{2} \int \sin 2x dx + c$$

$$y \sin x = -\frac{1}{4} \cos 2x + c$$

2. Solve $\frac{dy}{dx} + y \tan x = y^3 \sec x$

Step 1: Reduce it to an LDE

Divide by y^3 on both sides,

$$\frac{1}{y^3} \frac{dy}{dx} + \frac{1}{y^2} \tan x = \sec x \text{ ---- (1)}$$

$$\text{If } \frac{1}{y^2} = t \text{ then } -\frac{2}{y^3} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\text{Put } \frac{1}{y^2} = t, \frac{1}{y^3} \frac{dy}{dx} = -\frac{1}{2} \frac{dt}{dx} \text{ in (1)}$$

$$-\frac{1}{2} \frac{dt}{dx} + t \tan x = \sec x$$

Multiply by -2 on both sides,

$$\frac{dt}{dx} - 2t \tan x = -2 \sec x$$

This is an LDE in t with $P = -2 \tan x, Q = -2 \sec x$

Step 2: Solve reduced LDE

$$IF = e^{\int -2 \tan x dx} = e^{-2 \log \sec x} = \cos^2 x$$

General solution is $t \cdot IF = \int Q \cdot IF dx + c$

$$t \cos^2 x = \int -2 \sec x \cos^2 x dx + c$$

$$t \cos^2 x = -2 \sin x + c$$

$$\left(\frac{1}{y^2}\right) \cos^2 x = -2 \sin x + c$$

3. Solve $\frac{dy}{dx} + \frac{y}{x} = y^2x$ (May 22)

Step 1: Reduce it to an LDE

Divide by y^2 on both sides,

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{x} \left(\frac{1}{y} \right) = x \text{ ---- (1)}$$

$$\text{If } \frac{1}{y} = t \text{ then } -\frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\text{Put } \frac{1}{y} = t, \frac{1}{y^2} \frac{dy}{dx} = -\frac{dt}{dx} \text{ in (1)}$$

$$-\frac{dt}{dx} + \frac{t}{x} = x$$

Multiply by -1 on both sides,

$$\frac{dt}{dx} - \frac{t}{x} = -x$$

This is an LDE in t with $P = -1/x, Q = -x$

Step 2: Solve reduced LDE

$$IF = e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

General solution is $t \cdot IF = \int Q \cdot IF dx + c$

$$t \frac{1}{x} = \int -x \frac{1}{x} dx + c$$

$$t \frac{1}{x} = -x + c$$

$$\frac{1}{xy} = -x + c$$

4. $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$

Step 1: Reduce it to an LDE

Divide by $\cos y$ on both sides,

$$\sec y \tan y \frac{dy}{dx} + \sec y \tan x = \cos^2 x \text{ ---- (1)}$$

$$\text{If } \sec y = t \text{ then } \sec y \tan y \frac{dy}{dx} = \frac{dt}{dx}$$

$$(1) \Rightarrow \frac{dt}{dx} + t \tan x = \cos^2 x$$

This is an LDE in t with $P = \tan x, Q = \cos^2 x$

Step 2: Solve reduced LDE

$$IF = e^{\int P dx} = e^{\int \tan x dx} = \sec x$$

General solution is $t \cdot IF = \int Q \cdot IF dx + c$

$$t \sec x = \int \cos^2 x \sec x dx + c$$

$$\sec y \sec x = \sin x + c$$

5. Solve: $r \sin \theta - \cos \theta \frac{dr}{d\theta} = r^2$

Step 1: Reduce it to an LDE

Divide by r^2 on both sides,

$$-\frac{\cos \theta}{r^2} \frac{dr}{d\theta} + \frac{\sin \theta}{r} = 1 \text{ ---- (1)}$$

If $\frac{1}{r} = t$ then $-\frac{1}{r^2} \frac{dr}{d\theta} = \frac{dt}{d\theta}$

$$(1) \Rightarrow \cos \theta \frac{dt}{d\theta} + t \sin \theta = 1$$

Divide by $\cos \theta$ on both sides,

$$\frac{dt}{d\theta} + t \tan \theta = \sec \theta$$

This is an LDE in t with $P = \tan \theta, Q = \sec \theta$

Step 2: Solve reduced LDE

$$IF = e^{\int P d\theta} = e^{\int \tan \theta d\theta} = \sec \theta$$

General solution is $t \cdot IF = \int Q \cdot IF d\theta + c$

$$t \sec \theta = \int \sec \theta \sec \theta d\theta + c$$

$$\frac{1}{r} \sec \theta = \tan \theta + c$$

6. Solve: $\frac{dz}{dx} + \frac{z}{x} \log z = \frac{z}{x^2} (\log z)^2$

Step 1: Reduce it to an LDE

Divide by $z(\log z)^2$ on both sides,

$$\frac{1}{z(\log z)^2} \frac{dz}{dx} + \frac{1}{\log z} \left(\frac{1}{x} \right) = \frac{1}{x^2} \text{ ---- (1)}$$

If $\frac{1}{\log z} = t$ then $-\frac{1}{z(\log z)^2} \frac{dz}{dx} = \frac{dt}{dx}$

$$(1) \Rightarrow -\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

Multiply by -1 on both sides,

$$\frac{dt}{dx} - \frac{t}{x} = -\frac{1}{x^2}$$

This is an LDE in t with $P = -\frac{1}{x}, Q = -\frac{1}{x^2}$

Step 2: Solve reduced LDE

$$IF = e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

General solution is $t \cdot IF = \int Q \cdot IF dx + c$

$$t \frac{1}{x} = \int -\frac{1}{x^2} \cdot \frac{1}{x} dx + c$$

$$t \frac{1}{x} = \frac{x^{-2}}{2} + c$$

$$\frac{1}{x \log z} = \frac{1}{2x^2} + c$$

7. Solve: $x \frac{dy}{dx} + y = x^3 y^6$

Step 1: Reduce it to an LDE

Divide by x on both sides,

$$\frac{dy}{dx} + \left(\frac{1}{x}\right)y = x^2 y^6$$

Divide by y^6 on both sides,

$$\frac{1}{y^6} \frac{dy}{dx} + \left(\frac{1}{x}\right) \frac{1}{y^5} = x^2 \text{ ---- (1)}$$

$$\text{If } \frac{1}{y^5} = t \text{ then } -\frac{5}{y^6} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\text{Put } \frac{1}{y^5} = t, \frac{1}{y^6} \frac{dy}{dx} = -\frac{1}{5} \frac{dt}{dx} \text{ in (1)}$$

$$-\frac{1}{5} \frac{dt}{dx} + \frac{t}{x} = x^2$$

Multiply by -5 on both sides,

$$\frac{dt}{dx} - 5 \frac{t}{x} = -5x^2$$

This is an LDE in t with $P = -5/x, Q = -5x^2$

Step 2: Solve reduced LDE

$$IF = e^{\int -\frac{5}{x} dx} = e^{-5 \log x} = \frac{1}{x^5}$$

General solution is $t \cdot IF = \int Q \cdot IF dx + c$

$$t \frac{1}{x^5} = \int -5x^2 \frac{1}{x^5} dx + c$$

$$\frac{t}{x^5} = -5 \int x^{-3} dx$$

$$\frac{1}{x^5 y^5} = \frac{5}{2x^2} + c$$

8. Solve: $xy(1 + xy^2) \frac{dy}{dx} = 1$

Step 1: Reduce it to an LDE

$$xy + x^2y^3 = \frac{dx}{dy}$$

$$\frac{dx}{dy} - xy = x^2y^3$$

Divide by x^2 on both sides,

$$\frac{1}{x^2} \frac{dx}{dy} - y \left(\frac{1}{x} \right) = y^3$$

$$\text{If } \frac{1}{x} = t \text{ then } -\frac{1}{x^2} \frac{dx}{dy} = \frac{dt}{dy}$$

$$\text{Put } \frac{1}{x} = t, \frac{1}{x^2} \frac{dx}{dy} = -\frac{dt}{dy} \text{ in (1)}$$

$$-\frac{dt}{dy} - yt = y^3$$

Multiply by -1 on both sides,

$$\frac{dt}{dy} + yt = -y^3$$

This is an LDE in t with $P = y, Q = -y^3$

Step 2: Solve reduced LDE

$$IF = e^{\int y dy} = e^{\frac{y^2}{2}}$$

General solution is $t \cdot IF = \int Q \cdot IF dy + c$

$$te^{\frac{y^2}{2}} = \int -y^3 e^{\frac{y^2}{2}} dy + c$$

$$\text{Put } p = \frac{y^2}{2}, dp = y dy$$

$$te^p = \int -2pe^p dp + c$$

$$te^p = -2(pe^p - e^p) + c$$

$$\frac{1}{x} e^{\frac{y^2}{2}} = -2 \left(\frac{y^2}{2} e^{\frac{y^2}{2}} - e^{\frac{y^2}{2}} \right) + c$$

$$\frac{1}{x} e^{\frac{y^2}{2}} = (2 - y^2) e^{\frac{y^2}{2}} + c$$

9. Solve: $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$

Step 1: Reduce it to an LDE

Divide by $\cos^2 y$ on both sides,

$$\sec^2 y \frac{dy}{dx} + 2x \sec^2 y \sin y \cos y = x^3$$

$$\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$$

$$\text{If } \tan y = t \text{ then } \sec^2 y \frac{dy}{dx} = \frac{dt}{dx}$$

$$(1) \Rightarrow \frac{dt}{dx} + 2xt = x^3$$

This is an LDE in t with $P = 2x, Q = x^3$

Step 2: Solve reduced LDE

$$IF = e^{\int P dx} = e^{\int 2x dx} = e^{x^2}$$

General solution is $t \cdot IF = \int Q \cdot IF dx + c$

$$te^{x^2} = \int x^3 e^{x^2} dx + c, \text{ Put } p = x^2, dp = 2x dx$$

$$te^p = \frac{1}{2} \int p e^p dp + c$$

$$te^p = \frac{1}{2}(p - 1)e^p + c$$

$$(\tan y)e^{x^2} = \frac{1}{2}(x^2 - 1)e^{x^2} + c$$

10. Solve: $\frac{dy}{dx} = \frac{y}{x - \sqrt{xy}}$

Step 1: Reduce it to an LDE

$$\frac{x - \sqrt{xy}}{y} = \frac{dx}{dy}$$

$$\frac{dx}{dy} - \frac{x}{y} = -\sqrt{\frac{x}{y}}$$

Divide by $\sqrt{x}$ on both sides,

$$\frac{1}{\sqrt{x}} \frac{dx}{dy} - \sqrt{x} \left(\frac{1}{y} \right) = -\frac{1}{\sqrt{y}}$$

If $\sqrt{x} = t$ then $\frac{1}{2\sqrt{x}} \frac{dx}{dy} = \frac{dt}{dy}$

Put $\sqrt{x} = t$, $\frac{1}{\sqrt{x}} \frac{dx}{dy} = 2 \frac{dt}{dy}$ in (1)

$$2 \frac{dt}{dy} - \left(\frac{1}{y} \right) t = -\frac{1}{\sqrt{y}}$$

Divide by 2 on both sides,

$$\frac{dt}{dy} - \frac{1}{2y} t = -\frac{1}{2\sqrt{y}}$$

This is an LDE in t with $P = -\frac{1}{2y}$, $Q = -\frac{1}{2\sqrt{y}}$

Step 2: Solve reduced LDE

$$IF = e^{\int P dy} = e^{\int -\frac{1}{2y} dy} = \frac{1}{\sqrt{y}}$$

General solution is $t \cdot IF = \int Q \cdot IF dy + c$

$$t \frac{1}{\sqrt{y}} = \int -\frac{1}{2\sqrt{y}} \frac{1}{\sqrt{y}} dy + c$$

$$\sqrt{\frac{x}{y}} = \int -\frac{1}{2y} dy + c$$

$$\sqrt{\frac{x}{y}} = -\frac{1}{2} \log y + c$$

Home work:

1. $\frac{dy}{dx} = y \tan x - y^2 \sec x$. (P.T.U., 2005)	2. $r \sin \theta - \cos \theta \frac{dr}{d\theta} = r^2$. (V.T.U., 2005)
3. $2xy' = 10x^3y^2 + y$.	4. $(x^3y^2 + xy) dx = dy$. (B.P.T.U., 2005)
5. $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$. (Bhullai, 2005)	6. $x(x-y) dy + y^2 dx = 0$. (J.S.M., 2001)
7. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$. (Bhopal, 2009)	8. $e^x \left(\frac{dy}{dx} + 1 \right) = e^x$. (V.T.U., 2009)
9. $\sec^2 y \frac{dy}{dx} + x \tan y = x^2$.	10. $\tan y \frac{dy}{dx} + \tan x = \cos y \cos^2 x$. (Sambalpur, 2002)
11. $\frac{dy}{dx} = \frac{y}{x - \sqrt{xy}}$. (V.T.U., 2011)	12. $(y \log x - 2) y dx - x dy = 0$. (V.T.U., 2006)

3.2 Exact and reducible to exact differential equations

Exact differential equation:

- ❖ A differential equation of the form $M(x, y)dx + N(x, y)dy = 0$ is said to be exact if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

- ❖ General solution of an exact differential equation is

$$\int_{y=\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = 0$$

Reducible to exact differential equation:

- ❖ If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, given differential equation is not exact.
- ❖ Reduce it to an exact differential equation by multiplying I.F on both sides.
- ❖ If $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}$ is close to N then $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$. Now $I.F = e^{\int f(x) dx}$
- ❖ If $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}$ is close to M then $\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = g(y)$. Now $I.F = e^{\int g(y) dy}$

Problems:

1. Solve: $(x^2 + y^2 + x)dx + xy dy = 0$ (May 22)

$$(x^2 + y^2 + x)dx + xy dy = 0 \text{ ----- (1)}$$

$M = x^2 + y^2 + x$	$N = xy$
$\frac{\partial M}{\partial y} = 2y$	$\frac{\partial N}{\partial x} = y$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = y, \text{ close to } N.$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{xy} (y) = \frac{1}{x} = f(x) \text{ [say]}$$

$$I.F = e^{\int f(x) dx} = e^{\int \frac{1}{x} dx} = x$$

Multiply by x on both the sides of equation (1)

$$(x^3 + xy^2 + x^2)dx + x^2y dy = 0$$

This is an exact D.E.

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} (x^3 + xy^2 + x^2) dx + \int (0) dy = c$$

$$\frac{x^4}{4} + \frac{x^2y}{2} + \frac{x^3}{3} = c$$

2. Solve: $(4xy + 3y^2 - x) dx + x(x + 2y)dy = 0$

$$(4xy + 3y^2 - x) dx + (x^2 + 2xy)dy = 0 \text{ ----- (1)}$$

$M = 4xy + 3y^2 - x$	$N = x^2 + 2xy$
$\frac{\partial M}{\partial y} = 4x + 6y$	$\frac{\partial N}{\partial x} = 2x + 2y$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2x + 4y = 2(x + 2y), \text{ close to } N.$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{x^2 + 2xy} 2(x + 2y) = \frac{2}{x} = f(x) \text{ [say]}$$

$$I.F = e^{\int f(x) dx} = e^{\int \frac{2}{x} dx} = x^2$$

Multiply by x^2 on both the sides of equation (1)

$$(4x^3y + 3x^2y^2 - x^3) dx + (x^4 + 2x^3y)dy = 0$$

This is an exact D.E.

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} (4x^3y + 3x^2y^2 - x^3) dx + \int (0) dy = c$$

$$x^4y + x^3y^2 + \frac{x^4}{4} = c$$

3. Solve: $(xy^2 - e^{1/x^3}) dx - x^2y dy = 0$

$$(xy^2 - e^{1/x^3}) dx + (-x^2y)dy = 0 \text{ ----- (1)}$$

$M = xy^2 - e^{\frac{1}{x^3}}$	$N = -x^2y$
$\frac{\partial M}{\partial y} = 2xy$	$\frac{\partial N}{\partial x} = -2xy$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 4xy, \text{ close to N.}$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{-x^2y} (4xy) = -\frac{4}{x} = f(x) \text{ [say]}$$

$$I.F = e^{\int f(x) dx} = e^{-4 \int \frac{1}{x} dx} = x^{-4}$$

Multiply by x^{-4} on both the sides of equation (1)

$$x^{-4}(xy^2 - e^{x^{-3}}) dx + x^{-2}y dy = 0$$

This is an exact D.E.

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} (x^{-3}y^2 - x^{-4}e^{x^{-3}})dx + \int (0) dy = c$$

$$-\frac{1}{2}x^{-2}y^2 + \frac{1}{3}e^{x^{-3}} = c$$

4. Solve: $(x^2 + y^3 + 6x) dx + xy^2 dy = 0$

$$(x^2 + y^3 + 6x) dx + xy^2 dy = 0 \text{ ----- (1)}$$

$M = x^2 + y^3 + 6x$	$N = xy^2$
$\frac{\partial M}{\partial y} = 3y^2$	$\frac{\partial N}{\partial x} = y^2$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2y^2, \text{ close to } N.$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{xy^2} (2y^2) = \frac{2}{x} = f(x) \text{ [say]}$$

$$I.F = e^{\int f(x) dx} = e^{2 \int \frac{1}{x} dx} = x^2$$

Multiply by x^2 on both the sides of equation (1)

$$(x^{-3}y^2 - x^{-4}e^{1/x^3}) dx - x^{-2}y dy = 0$$

This is an exact D.E.

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} (x^{-3}y^2 - x^{-4}e^{1/x^3}) dx + \int (0) dy = c$$

$$-\frac{y^2}{2x^2} + \frac{e^{1/x^3}}{3} = c$$

5. Solve: $(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$

$$(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0 \text{ ----- (1)}$$

$M = y^4 + 2y$	$N = xy^3 + 2y^4 - 4x$
$\frac{\partial M}{\partial y} = 4y^3 + 2$	$\frac{\partial N}{\partial x} = y^3 - 4$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 3y^3 + 6 = 3(y^3 + 2), \text{ close to M.}$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{3}{y^4 + 2y} (y^3 + 2) = \frac{3}{y} = g(y) \text{ [say]}$$

$$I.F = e^{-\int g(y) dy} = e^{-3 \int \frac{1}{y} dy} = y^{-3}$$

Multiply by y^3 on both the sides of equation (1)

$$y^{-3}(y^4 + 2y) dx + y^{-3}(xy^3 + 2y^4 - 4x) dy = 0$$

This is an exact D.E.

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} (y + 2y^{-2}) dx + \int (2y) dy = c$$

$$xy + \frac{2x}{y^2} + y^2 = c$$

6. Solve: $(3x^2y^4 + 2xy) dx + (2x^3y^3 - x^2) dy = 0$

$$(3x^2y^4 + 2xy) dx + (2x^3y^3 - x^2)dy = 0 \text{ ----- (1)}$$

$M = 3x^2y^4 + 2xy$	$N = 2x^3y^3 - x^2$
$\frac{\partial M}{\partial y} = 12x^2y^3 + 2x$	$\frac{\partial N}{\partial x} = 6x^2y^3 - 2x$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 6x^2y^3 + 4x = 2(3x^2y^3 + 2x), \text{ close to M.}$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{2}{3x^2y^4 + 2xy} (3x^2y^3 + 2x) = \frac{2}{y} = g(y) \text{ [say]}$$

$$I.F = e^{-\int g(y) dy} = e^{-2 \int \frac{1}{y} dy} = y^{-2}$$

Multiply by y^{-2} on both the sides of equation (1)

$$y^{-2}(3x^2y^4 + 2xy) dx + y^{-2}(2x^3y^3 - x^2)dy = 0$$

This is an exact D.E.

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} \left(3x^2y^2 + \frac{2x}{y} \right) dx + \int (0) dy = c$$

$$x^3y^2 + \frac{x^2}{y} = c$$

7. Solve: $(xy^3 + y) dx + 2(x^2y^2 + x + y^4) dy = 0$

$$(xy^3 + y) dx + 2(x^2y^2 + x + y^4)dy = 0 \text{ ----- (1)}$$

$M = xy^3 + y$	$N = 2(x^2y^2 + x + y^4)$
$\frac{\partial M}{\partial y} = 3xy^2 + 1$	$\frac{\partial N}{\partial x} = 2(2xy^2 + 1) = 4xy^2 + 2$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = -xy^2 - 1 = -(xy^2 + 1), \text{ close to M.}$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = -\frac{xy^2+1}{xy^3+y} = -\frac{1}{y} = g(y) \text{ [say]}$$

$$I.F = e^{-\int g(y) dy} = e^{\int \frac{1}{y} dx} = y$$

Multiply by y on both the sides of equation (1)

$$y(xy^3 + y) dx + 2y(x^2y^2 + x + y^4)dy = 0 \text{ This is an exact D.E.}$$

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} (xy^4 + y^2)dx + \int (2y^5) dy = c$$

$$\frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = c$$

8. Solve: $(y \log y) dx + (x - \log y) dy = 0$

$$(y \log y) dx + (x - \log y) dy = 0 \text{ ----- (1)}$$

$M = y \log y$	$N = x - \log y$
$\frac{\partial M}{\partial y} = y \left(\frac{1}{y} \right) + \log y$	$\frac{\partial N}{\partial x} = 1$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 1 + \log y - 1 = \log y, \text{ close to M.}$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{y \log y} (\log y) = \frac{1}{y} = g(y) \text{ [say]}$$

$$I.F = e^{-\int g(y) dy} = e^{-\int \frac{1}{y} dx} = \frac{1}{y}$$

Multiply by y on both the sides of equation (1)

$$\frac{1}{y} (y \log y) dx + \frac{1}{y} (x - \log y) dy = 0 \text{ This is an exact D.E.}$$

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} \log y dx + \int \left(\frac{1}{y} \log y \right) dy = c$$

$$x \log y + \frac{(\log y)^2}{2} = c$$

9. Solve: $y(x + y + 1) dx + x(x + 3y + 2) dy = 0$

$$(xy + y^2 + y) dx + (x^2 + 3xy + 2x)dy = 0 \text{ ----- (1)}$$

$M = xy + y^2 + y$	$N = x^2 + 3xy + 2x$
$\frac{\partial M}{\partial y} = x + 2y + 1$	$\frac{\partial N}{\partial x} = 2x + 3y + 2$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = -x - y - 1 = -(x + y + 1), \text{ close to M.}$$

$$\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = -\frac{1}{xy+y^2+y} (x + y + 1) = -\frac{1}{y} = g(y) \text{ [say]}$$

$$I.F = e^{-\int g(y) dy} = e^{\int \frac{1}{y} dx} = y$$

Multiply by y on both the sides of equation (1)

$$y(xy + y^2 + y) dx + y(x^2 + 3xy + 2x)dy = 0$$

This is an exact D.E.

General solution is

$$\int_{y-\text{constant}} M dx + \int (\text{Terms of } N \text{ not containing } x) dy = c$$

$$\int_{y-\text{constant}} (xy^2 + y^3 + y^2)dx + \int (0) dy = c$$

$$\frac{x^2 y^2}{2} + xy^3 + xy^2 = c$$

10. Solve: $2y \, dx + (2x \log x - xy) \, dy = 0$

$$(2y) \, dx + (2x \log x - xy) \, dy = 0 \text{ ----- (1)}$$

$M = 2y$	$N = 2x \log x - xy$
$\frac{\partial M}{\partial y} = 2$	$\frac{\partial N}{\partial x} = 2x \left(\frac{1}{x}\right) + 2 \log x - y$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, this is not an exact D.E.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = -2 \log x + y = -(2 \log x - y), \text{ close to } N.$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{2x \log x - xy} (-2 \log x + y) = -\frac{1}{x} = f(x) \text{ [say]}$$

$$I.F = e^{\int f(x) \, dx} = e^{-\int \frac{1}{x} \, dx} = \frac{1}{x}$$

Multiply by y on both the sides of equation (1)

$$\frac{1}{x} (2y) \, dx + \frac{1}{x} (2x \log x - xy) \, dy = 0 \text{ This is an exact D.E.}$$

General solution is

$$\int_{y-\text{constant}} M \, dx + \int (\text{Terms of } N \text{ not containing } x) \, dy = c$$

$$\int_{y-\text{constant}} \frac{2y}{x} \, dx + \int (-y) \, dy = c$$

$$2y \log x - \frac{y^2}{2} = c$$

3.3 Orthogonal Trajectory

Definition: Two families of curves such that every member of either family cuts each member of the other family at right angles are called orthogonal trajectories.

Example: Family of circles $x^2 + y^2 = a^2$ is the orthogonal trajectories to the family of straight lines $y = mx + c$. Where a and m are arbitrary constants.

Working rule to find the orthogonal trajectories of the family of curves $f(x, y, c) = 0$:

- ❖ Form the differential equation by eliminating arbitrary constant c .
- ❖ Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$. [$\tan(90^\circ + \theta) = -\cot \theta$]
- ❖ Solve the modified differential equation.

Working rule to find the orthogonal trajectories of the family of curves $f(r, \theta, c) = 0$:

- ❖ Form the differential equation by eliminating arbitrary constant c .
- ❖ Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$. [$\tan(90^\circ + \phi) = -\cot \phi$]
- ❖ Solve the modified differential equation.

Problems:

1. Find the orthogonal trajectories of the family of parabolas $y^2 = 4ax$.

Consider $y^2 = 4ax$ ----- (1)

Differentiate w.r.to x ,

$$2y \frac{dy}{dx} = 4a$$

Substitute in (1),

$$y^2 = 2xy \frac{dy}{dx}$$

$$y = 2x \frac{dy}{dx}$$

Replace $\frac{dy}{dx} = -\frac{dx}{dy}$

$$y = -2x \frac{dx}{dy}$$

$$y dy = -2x dx$$

On integrating,

$$\frac{y^2}{2} = -x^2 + c$$

$$2x^2 + y^2 = k$$

This is the family of orthogonal trajectories of (1).

2. Find the orthogonal trajectories of the family of circles $x^2 + y^2 = a^2$.

Consider $x^2 + y^2 = a^2$ ----- (1)

Differentiate w.r.to x ,

$$2x + 2y \frac{dy}{dx} = 0$$

$$y \frac{dy}{dx} = -x$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$

$$y \left(-\frac{dx}{dy} \right) = -x$$

$$\frac{1}{x} dx = \frac{1}{y} dy$$

On integrating,

$$\log x = \log y + \log c$$

$$x = yc$$

This is the family of orthogonal trajectories of (1).

3. Find the orthogonal trajectories of the family of curves $y^2 = c x^3$.

Consider $y^2 = c x^3$ ----- (1)

Differentiate w.r.to x ,

$$2y \frac{dy}{dx} = 3c x^2$$

$$\times x \Rightarrow 2xy \frac{dy}{dx} = 3c x^3$$

By (1), $2xy \frac{dy}{dx} = 3y^2$

$$2x \frac{dy}{dx} = 3y$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$

$$2x \left(-\frac{dx}{dy} \right) = 3y$$

$$-2x dx = 3y dy$$

On integrating,

$$-x^2 = \frac{3y^2}{2} + c$$

$$2x^2 + 3y^2 = k$$

This is the family of orthogonal trajectories of (1).

4. Find the orthogonal trajectories of the family of curves $x^{2/3} + y^{2/3} = a^{2/3}$.

Consider $x^{2/3} + y^{2/3} = a^{2/3}$ ----- (1)

Differentiate w.r.to x ,

$$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}}\frac{dy}{dx} = 0$$

$$\times \frac{3}{2} \Rightarrow x^{-\frac{1}{3}} + y^{-\frac{1}{3}}\frac{dy}{dx} = 0$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$

$$x^{-\frac{1}{3}} + y^{-\frac{1}{3}}\left(-\frac{dx}{dy}\right) = 0$$

$$y^{-\frac{1}{3}}\left(-\frac{dx}{dy}\right) = -x^{-\frac{1}{3}}$$

$$x^{\frac{1}{3}}dx = y^{\frac{1}{3}}dy$$

On integrating,

$$x^{4/3} = y^{4/3} + c \Rightarrow x^{4/3} - y^{4/3} = c$$

This is the family of orthogonal trajectories of (1).

5. Show that the family of parabolas $y^2 = 4a(x + a)$ is self-orthogonal.

$$y^2 = 4a(x + a)$$

Diff. w. r. to x ,

$$2y\frac{dy}{dx} = 4a$$

By substituting in (1),

$$y^2 = 2y\frac{dy}{dx}\left(x + \frac{y}{2}\frac{dy}{dx}\right)$$

$$y^2 = 2xy\frac{dy}{dx} + y^2\left(\frac{dy}{dx}\right)^2$$

$$y = 2x\frac{dy}{dx} + y\left(\frac{dy}{dx}\right)^2 \text{ ----- (1)}$$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$

$$y = 2x\left(-\frac{dx}{dy}\right) + y\left(-\frac{dx}{dy}\right)^2$$

$$y = -2x\left(\frac{dx}{dy}\right) + y\left(\frac{dx}{dy}\right)^2$$

$$y\left(\frac{dy}{dx}\right)^2 = -2x\left(\frac{dy}{dx}\right) + y$$

$$y = y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) \text{ ----- (2)}$$

Since (1) = (2), The given family of parabolas is self-orthogonal.

6. Find the orthogonal trajectories of the family of curves $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$, where λ is the parameter. (May 22)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1 \text{ ----- (1)}$$

Diff. w.r.to x ,

$$\frac{2x}{a^2} + \frac{2y}{b^2 + \lambda} \frac{dy}{dx} = 0$$

$$\frac{y}{b^2 + \lambda} \frac{dy}{dx} = -\frac{x}{a^2}$$

$$\frac{y}{b^2 + \lambda} = -\frac{x}{a^2} \left(\frac{dx}{dy} \right)$$

Substitute in (1),

$$\frac{x^2}{a^2} - \frac{xy}{a^2} \left(\frac{dx}{dy} \right) = 1$$

$$x^2 - xy \left(\frac{dx}{dy} \right) = a^2$$

$$x^2 - a^2 = xy \left(\frac{dx}{dy} \right)$$

$$\frac{dy}{dx} = \frac{xy}{x^2 - a^2}$$

Replace $\frac{dy}{dx} = -\frac{dx}{dy}$

$$-\frac{dx}{dy} = \frac{xy}{x^2 - a^2}$$

$$\frac{a^2 - x^2}{x} dx = y dy$$

$$\left(\frac{a^2}{x} - x \right) dx = y dy$$

$$a^2 \log x - \frac{x^2}{2} = \frac{y^2}{2} + c$$

$$x^2 + y^2 = 2a^2 \log x + k$$

This is the family orthogonal trajectories of (1).

7. Find the orthogonal trajectories of the family of curves $x^3 - 3xy^2 = c$
 $x^3 - 3xy^2 = c$ ----- (1)

Diff. w.r.to x ,

$$3x^2 - 3y^2 - 6xy \frac{dy}{dx} = 0$$

$$x^2 - y^2 = 2xy \frac{dy}{dx}$$

Replace $\frac{dy}{dx} = -\frac{dx}{dy}$

$$x^2 - y^2 = 2xy \left(-\frac{dx}{dy} \right)$$

$$\frac{dy}{dx} = \frac{2xy}{y^2 - x^2}$$

$$\begin{aligned}\text{Put } y &= vx, \frac{dy}{dx} = v + x \frac{dv}{dx} \\ v + x \frac{dv}{dx} &= \frac{2x(vx)}{(vx)^2 - x^2} = \frac{2v}{v^2 - 1} \\ x \frac{dv}{dx} &= \frac{2v}{v^2 - 1} - v = \frac{2v - v^3 + v}{v^2 - 1} = \frac{3v - v^3}{v^2 - 1} \\ \frac{v^2 - 1}{3v - v^3} dv &= \frac{1}{x} dx \\ -\frac{1}{3} \left(\frac{3 - 3v^2}{3v - v^3} \right) dv &= \frac{1}{x} dx\end{aligned}$$

On integrating,

$$\begin{aligned}-\frac{1}{3} \log(3v - v^3) &= \log x + \log c \\ \log(3v - v^3) &= -3 \log x - 3 \log c \\ x^3(3v - v^3) &= k \\ x^3 \left(3 \left(\frac{y}{x} \right) - \frac{y^3}{x^3} \right) &= k\end{aligned}$$

$3x^2y - y^3 = k$. This is the family orthogonal trajectories of (1).

8. Find the orthogonal trajectories of the family of curves $r^n = a^n \cos n\theta$.

$$\begin{aligned}r^n &= a^n \cos n\theta \\ \log r^n &= \log a^n \cos n\theta \\ n \log r &= \log a^n + \log \cos n\theta \\ \frac{n}{r} \frac{dr}{d\theta} &= -n \frac{\sin n\theta}{\cos n\theta} \\ \frac{1}{r} \frac{dr}{d\theta} &= -\frac{\sin n\theta}{\cos n\theta}\end{aligned}$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$\begin{aligned}-r \frac{d\theta}{dr} &= -\tan n\theta \\ \cot n\theta d\theta &= \frac{1}{r} dr\end{aligned}$$

On integrating,

$$\begin{aligned}\frac{1}{n} \log \sin n\theta &= \log r + \log c \\ \log \sin n\theta &= n \log rc \\ \sin n\theta &= r^n c^n \\ r^n &= k \sin n\theta\end{aligned}$$

This is the required O.T.

9. Find the orthogonal trajectories of the family of curves $r^n \cos n\theta = a^n$.

$$r^n \cos n\theta = a^n$$

$$\log(r^n \cos n\theta) = \log a^n$$

$$n \log r + \log \cos n\theta = \log a^n$$

$$\frac{n}{r} \frac{dr}{d\theta} - n \frac{\sin n\theta}{\cos n\theta} = 0$$

$$\frac{1}{r} \frac{dr}{d\theta} = \tan n\theta$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$-r \frac{d\theta}{dr} = \tan n\theta$$

$$\cot n\theta d\theta = -\frac{1}{r} dr$$

On integrating,

$$\frac{1}{n} \log \sin n\theta = -\log r + \log c$$

$$\log \sin n\theta = n \log \frac{c}{r}$$

$$\sin n\theta = \frac{c^n}{r^n}$$

$r^n \sin n\theta = k$. This is the required O.T.

10. Find the orthogonal trajectories of the family of curves $r = 2a \cos \theta$.

$$r = 2a \cos \theta$$

$$\log r = \log(2a \cos \theta)$$

$$\log r = \log 2a + \log \cos \theta$$

$$\frac{1}{r} \frac{dr}{d\theta} = -\frac{\sin \theta}{\cos \theta}$$

$$\frac{1}{r} \frac{dr}{d\theta} = -\tan \theta$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$-r \frac{d\theta}{dr} = -\tan \theta$$

$$\cot \theta d\theta = \frac{1}{r} dr$$

On integrating,

$$\log \sin \theta = \log r + \log c$$

$$\log \sin \theta = \log cr$$

$$\sin \theta = cr$$

$$r = k \sin \theta$$

This is the required O.T.

11. Find the orthogonal trajectories of the family of curves $r^n = a^n \sin n\theta$.

$$r^n = a^n \sin n\theta$$

$$\log r^n = \log a^n \sin n\theta$$

$$n \log r = \log a^n + \log \sin n\theta$$

$$\frac{n}{r} \frac{dr}{d\theta} = n \frac{\cos n\theta}{\sin n\theta}$$

$$\frac{1}{r} \frac{dr}{d\theta} = \cot n\theta$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$-r \frac{d\theta}{dr} = \cot n\theta$$

$$\tan n\theta d\theta = -\frac{1}{r} dr$$

On integrating,

$$\frac{1}{n} \log \sec n\theta = -\log r + \log c$$

$$\log \sec n\theta = n \log \frac{c}{r}$$

$$\sec n\theta = \frac{c^n}{r^n}$$

$r^n = k \cos n\theta$. This is the required O.T.

12. Find the orthogonal trajectories of the family of curves $r = a(1 - \cos \theta)$.

$$r = a(1 - \cos \theta)$$

$$\log r = \log a + \log(1 - \cos \theta)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta}$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$-r \frac{d\theta}{dr} = \frac{\sin \theta}{1 - \cos \theta}$$

$$-\frac{1 - \cos \theta}{\sin \theta} d\theta = \frac{1}{r} dr$$

$$-\frac{1 - \cos^2 \theta}{\sin \theta (1 + \cos \theta)} d\theta = \frac{1}{r} dr$$

$$-\frac{\sin^2 \theta}{\sin \theta (1 + \cos \theta)} d\theta = \frac{1}{r} dr$$

$$-\frac{\sin \theta}{(1 + \cos \theta)} d\theta = \frac{1}{r} dr$$

On integrating,

$$\log(1 + \cos \theta) = \log r + \log c$$

$$\log(1 + \cos \theta) = \log cr$$

$1 + \cos \theta = cr$ (or) $r = k(1 + \cos \theta)$. This is the required O.T.

13. Find the orthogonal trajectories of the family of curves $r = a(1 + \sin \theta)$.

$$r = a(1 + \sin \theta)$$

$$\log r = \log a + \log(1 + \sin \theta)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\cos \theta}{1 + \sin \theta}$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$-r \frac{d\theta}{dr} = \frac{\cos \theta}{1 + \sin \theta}$$

$$-\frac{1 + \sin \theta}{\cos \theta} d\theta = \frac{1}{r} dr$$

$$-\frac{1 - \sin^2 \theta}{\cos \theta (1 - \sin \theta)} d\theta = \frac{1}{r} dr$$

$$-\frac{\cos \theta}{1 - \sin \theta} d\theta = \frac{1}{r} dr$$

On integrating,

$$\log(1 - \sin \theta) = \log r + \log c$$

$$1 - \sin \theta = cr$$

$r = k(1 - \sin \theta)$. This is the required O.T.

14. Find the orthogonal trajectories of the family of curves $r = 2a(\cos \theta + \sin \theta)$.

$$r = 2a(\cos \theta + \sin \theta)$$

$$\log r = \log 2a + \log(\cos \theta + \sin \theta)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$-r \frac{d\theta}{dr} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$$

$$-\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} d\theta = \frac{1}{r} dr$$

On integrating,

$$\log(\cos \theta - \sin \theta) = \log r + \log c$$

$$\log(\cos \theta - \sin \theta) = \log cr$$

$$\cos \theta - \sin \theta = cr$$

$$r = k(\cos \theta - \sin \theta)$$

This is the required O.T.

15. Find the orthogonal trajectories of the family of curves $r = 4a(\sec \theta + \tan \theta)$.

$$r = 4a(\sec \theta + \tan \theta)$$

$$\log r = \log 4a + \log(\sec \theta + \tan \theta)$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\sec \theta \tan \theta + \sec^2 \theta}{\sec \theta + \tan \theta}$$

$$\frac{1}{r} \frac{dr}{d\theta} = \sec \theta$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$-r \frac{d\theta}{dr} = \sec \theta$$

$$-\cos \theta d\theta = \frac{1}{r} dr$$

On integrating,

$$-\sin \theta = \log r + \log c$$

$$\log rc = -\sin \theta$$

$$rc = e^{-\sin \theta}$$

$$re^{\sin \theta} = k$$

This is the required O.T.

16. Prove that the orthogonal trajectories of the family of curves $\frac{2a}{r} = 1 - \cos \theta$ is

$$\frac{2b}{r} = 1 + \cos \theta.$$

$$\frac{2a}{r} = 1 - \cos \theta$$

$$\log \frac{2a}{r} = \log(1 - \cos \theta)$$

$$\log 2a - \log r = \log(1 - \cos \theta)$$

$$-\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta}$$

Replace $\frac{1}{r} \frac{dr}{d\theta}$ by $-r \frac{d\theta}{dr}$

$$r \frac{d\theta}{dr} = \frac{\sin \theta}{1 - \cos \theta}$$

$$\frac{1 - \cos \theta}{\sin \theta} d\theta = \frac{1}{r} dr$$

$$\frac{1 - \cos^2 \theta}{\sin \theta (1 + \cos \theta)} d\theta = \frac{1}{r} dr$$

$$\frac{\sin^2 \theta}{\sin \theta (1 + \cos \theta)} d\theta = \frac{1}{r} dr$$

$$\frac{\sin \theta}{(1 + \cos \theta)} d\theta = \frac{1}{r} dr$$

On integrating,

$$-\log(1 + \cos \theta) = \log r + \log c$$

$$\log 2b = \log r + \log(1 + \cos \theta)$$

$$\log \frac{2b}{r} = \log(1 + \cos \theta)$$

$$\frac{2b}{r} = 1 + \cos \theta$$

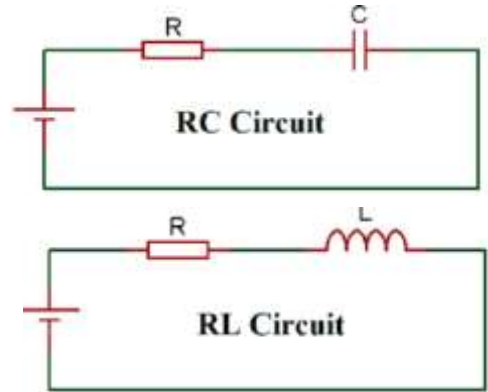
This is the required O.T.

3.4 RL and RC circuits

Introduction:

$$(i) I = \frac{dQ}{dt}$$

Notation	Terminology	Unit
L	Inductance	Henry
C	Capacitance	Farad
R	Resistance	Ohms
E	Electro motive force (e.m.f.)	Volts
I	Current	Amperes
Q	Charge	Coloumb



(ii) Voltage drop across resistance $R = RI$

(iii) Voltage drop across inductance $L = L \frac{dI}{dt}$

(iv) Voltage drop across capacitance $C = \frac{Q}{C}$

By Kirchhoff's law, $L \frac{dI}{dt} + RI = E$ in LR circuit.

By Kirchhoff's law, $RI + \frac{Q}{C} = E$ in RC circuit.

1. When a switch is closed in a circuit containing a battery E , a resistance R and an inductance L , the current I builds up at a rate given by $L \frac{di}{dt} + Ri = E$. Find i as a function of t . How long will it be, before the current has reached one-half its final value if $E = 6$ volts, $R = 100$ ohms and $L = 0.1$ henry?

To find I:

$$L \frac{dI}{dt} + RI = E, \text{ Put } E = 6, R = 100, L = 0.1$$

$$0.1 \frac{dI}{dt} + 100I = 6$$

$$\frac{dI}{dt} + 1000I = 60$$

$$\frac{dI}{60 - 1000I} = dt$$

$$-0.001 \log(60 - 1000I) = t + c$$

$$\log(60 - 1000I) = -1000t + c'$$

$$60 - 1000I = ke^{-1000t}$$

When $t = 0, i = 0$. so, $k = 60$

$$60 - 1000I = 60e^{-1000t}$$

$$1000I = 60 - 60e^{-1000t}$$

$$I = 0.06(1 - e^{-1000t})$$

To find t when I reaches max. value of I/2 :

When t is max, I is max. So, Max. value of $I = 0.06$

$$\text{Max. value of } \frac{I}{2} = 0.03$$

$$0.03 = 0.06(1 - e^{-1000t})$$

$$e^{1000t} = 2$$

$$t = 0.0006931 \text{ sec}$$

2. When a Resistance R ohms is connected in series with an inductance L henries with an e.m.f. of E volts, the current i amperes at time t is given by $L \frac{di}{dt} + Ri = E$. If $E = 10 \sin t$ volts and $i = 0$ when $t = 0$, find i as a function of t .

$$L \frac{di}{dt} + Ri = E, \text{ Put } E = 10 \sin t, i = 0, t = 0$$

$$L \frac{di}{dt} + Ri = 10 \sin t$$

$$\frac{di}{dt} + \frac{R}{L}i = 10 \sin t$$

$$i \cdot e^{\int \frac{R}{L} dt} = \int 10 \sin t \cdot e^{\int \frac{R}{L} dt} dt + c$$

$$i \cdot e^{\frac{Rt}{L}} = \int 10 \sin t \cdot e^{\frac{Rt}{L}} dt + c$$

$$i \cdot e^{\frac{Rt}{L}} = \frac{10e^{\frac{Rt}{L}}}{\frac{R^2}{L^2} + 1} \left(\frac{R}{L} \sin t - \cos t \right) + c$$

$$i = \frac{10L}{R^2 + L^2} (R \sin t - L \cos t) + ce^{-\frac{Rt}{L}}$$

By data, at $t = 0, i = 0$.

$$0 = \frac{10L}{L^2 + R^2} (-L) + c$$

Therefore, $c = \frac{10L^2}{L^2 + R^2}$

$$i = \frac{10L}{R^2 + L^2} \left(R \sin t - L \cos t + Le^{-\frac{Rt}{L}} \right)$$

3. A resistance of $100\ \Omega$, an inductance of 0.5 henry are connected in series with a battery of 20 volts. Find the current in the circuit at $t = 0.5$ secs. If $i = 0$ at $t = 0$.

$$L \frac{di}{dt} + Ri = E, \text{ Put } E = 20, R = 100, L = 0.5$$

$$0.5 \frac{di}{dt} + 100i = 20$$

$$\frac{di}{dt} + 200i = 40$$

$$\frac{di}{40 - 200i} = dt$$

$$-0.005 \log(40 - 200i) = t + c$$

$$\log(40 - 200i) = -200t + c'$$

$$40 - 200i = de^{-200t}$$

$$i = 0 \text{ at } t = 0.$$

$$40 = d$$

$$40 - 200i = 40e^{-200t}$$

$$1 - 5i = e^{-200t}$$

$$\text{At } t = 0.5$$

$$1 - 5i = e^{-100}$$

$$i = \frac{1 - e^{-100}}{5}$$

4. Find the current at any time $t > 0$, in a circuit having in series a constant electromotive force $40V$, a resistor 10Ω , an inductor $0.2H$ given that initial current is zero.

$$\text{By data, } E = 40, R = 10, L = 0.2$$

$$\text{By Kirchoff's law, } L \frac{dI}{dt} + RI = E$$

$$\text{On substituting, } 0.2 \frac{dI}{dt} + 10I = 40$$

$$\text{Therefore, } \frac{dI}{dt} + 50I = 200$$

Solution is given by

$$I \cdot e^{50t} = \int 200 e^{50t} dt + c$$

$$I = 4 + ce^{-50t}$$

$$\text{By data, at } t = 0, I = 0.$$

$$0 = 4 + c, \quad c = -4$$

$$\text{Therefore, } I = 4 - 4e^{-50t} = 4(1 - e^{-50t})$$

5. A generator having e.m.f. 100 volts is connected in series with a 10 ohms resistor and an inductor of 2 henries. If the switch is closed at a time $t = 0$, determine the current at time $t > 0$.

By data, $E = 100, R = 10, L = 2$

By Kirchoff's law, $L \frac{dI}{dt} + RI = E$

On substituting, $2 \frac{dI}{dt} + 10I = 100$

Therefore, $\frac{dI}{dt} + 5I = 50$

Solution is given by

$$I \cdot e^{5t} = \int 50 e^{5t} dt + c$$

$$I \cdot e^{5t} = 10e^{5t} + c$$

$$I = 10 + ce^{-5t}$$

By data, at $t = 0, I = 0$.

$$0 = 10 + c, \quad c = -10$$

Therefore, $I = 10 - 10e^{-5t} = 10(1 - e^{-5t})$

6. A decaying e.m.f. $E = 200e^{-5t}$ is connected in series with a 20 ohm resistor and 0.01 farad capacitor. Find the charge and current at any time assuming $Q = 0$ at $t = 0$. Show that the charge reaches the maximum. Calculate it and find time when it is reached.

By data, $E = 200e^{-5t}$, $R = 20$, $C = 0.01$

By Kirchoff's law, $RI + \frac{Q}{C} = E$

On substituting, $20I + \frac{Q}{0.01} = 200e^{-5t}$

Therefore, $\frac{dQ}{dt} + 5Q = 10e^{-5t}$

Solution is given by

$$Q \cdot e^{5t} = \int 10e^{-5t} e^{5t} dt + c$$

$$Q \cdot e^{5t} = 10t + c$$

By data, at $t = 0$, $Q = 0$.

$$0 = 0 + c, \quad c = 0$$

Therefore, $Q \cdot e^{5t} = 10t$

$$Q = 10t e^{-5t}$$

$$\frac{dQ}{dt} = 10(e^{-5t} - 5te^{-5t}) = 10(1 - 5t)e^{-5t}$$

Q is maximum when $\frac{dQ}{dt} = 0$

$$10(1 - 5t)e^{-5t} = 0$$

$$t = \frac{1}{5}.$$

Maximum value of $Q = 10\left(\frac{1}{5}\right)e^{-1} = \frac{2}{e}$

3.5 Non-linear differential equations

Introduction: Product of variables and their first order derivatives are allowed in the non-linear differential equations.

Problems:

1. Solve: $p^2 + p(x + y) + xy = 0$

$$(p + x)(p + y) = 0$$

$p = -x$	$p = -y$
$\frac{dy}{dx} = -x$	$\frac{dy}{dx} = -y$
$dy = -x dx$	$\frac{1}{y} dy = -dx$
$\frac{x^2}{2} + y - c = 0$	$x + \log y - c = 0$

Therefore, the general solution is

$$\left(\frac{x^2}{2} + y - c\right)(x + \log y - c) = 0$$

2. Solve: $p^2 + 2p \cosh x + 1 = 0$

$$p^2 + p(e^x + e^{-x}) + 1 = 0$$

$$p(p + e^x) + e^{-x}(p + e^x) = 0$$

$$(p + e^x)(p + e^{-x}) = 0$$

$p = -e^x$	$p = -e^{-x}$
$\frac{dy}{dx} = -e^x$	$\frac{dy}{dx} = -e^{-x}$
$dy = -e^x dx$	$dy = -e^{-x} dx$
$e^x + y - c = 0$	$-e^{-x} + y - c = 0$

Therefore, general solution is

$$(y + e^x - c)(y - e^{-x} - c) = 0$$

3. Solve: $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$

$$xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$$

$$xyp^2 + 3x^2p - 2y^2p - 6xy = 0$$

$$xp(yp + 3x) - 2y(yp + 3x) = 0$$

$$(xp - 2y)(yp + 3x) = 0$$

$xp = 2y$	$yp = -3x$
$x \frac{dy}{dx} = 2y$	$y \frac{dy}{dx} = -3x$
$\frac{1}{y} dy = \frac{2}{x} dx$	$y dy = -3x dx$
$\log y = 2 \log x + c$	$\frac{y^2}{2} + \frac{3x^2}{2} - c = 0$
$\log y = \log cx^2$	$y^2 + 3x^2 - 2c = 0$
$y = cx^2$	

Therefore, general solution is

$$(y - cx^2)(y^2 + 3x^2 - 2c) = 0$$

4. Solve: $p(p + y) = x(x + y)$

$$p^2 + py - x^2 - xy = 0$$

$$(p^2 - x^2) + y(p - x) = 0$$

$$(p - x)(p + x + y) = 0$$

$p = x$	$p = -x - y$
$\frac{dy}{dx} = x$	$\frac{dy}{dx} = -x - y$
$dy = x dx$	$\frac{dy}{dx} + y = -x$
On integrating,	This is an L.D.E. Solution is
$y = \frac{x^2}{2} + c$	$ye^x = \int -x \cdot e^x dx + c$
$y - \frac{x^2}{2} - c = 0$	$ye^x = -(xe^x - e^x) + c$
	$e^x(x + y - 1) - c = 0$

Therefore, general solution is

$$\left[y - \frac{x^2}{2} - c\right] [e^x(x + y - 1) - c] = 0$$

5. Solve: $p^2 + 2py \cot x = y^2$

$$p^2 + 2py \cot x - y^2 = 0$$

$$(p + y \cot x)^2 - y^2 - y^2 \cot^2 x = 0$$

$$(p + y \cot x)^2 - y^2 \operatorname{cosec}^2 x = 0$$

$$(p + y \cot x + y \operatorname{cosec} x)(p + y \cot x - y \operatorname{cosec} x) = 0$$

$p + y \cot x + y \operatorname{cosec} x = 0$ $\frac{dy}{y} = (-\operatorname{cosec} x - \cot x) dx$ $\frac{dy}{y} = -\left(\frac{1+\cos x}{\sin x}\right) dx$ $\frac{dy}{y} = -\left(\frac{1-\cos^2 x}{\sin x(1-\cos x)}\right) dx$ $\frac{dy}{y} = -\left(\frac{\sin x}{1-\cos x}\right) dx$ On integrating, $\log y = -\log(1 - \cos x) + \log c$ $y(1 - \cos x) - c = 0$	$p + y \cot x - y \operatorname{cosec} x = 0$ $\frac{dy}{y} = (\operatorname{cosec} x - \cot x) dx$ $\frac{dy}{y} = \left(\frac{1-\cos x}{\sin x}\right) dx$ $\frac{dy}{y} = \left(\frac{1-\cos^2 x}{\sin x(1+\cos x)}\right) dx$ $\frac{dy}{y} = \left(\frac{\sin x}{1+\cos x}\right) dx$ On integrating, $\log y = -\log(1 + \cos x) + \log c$ $y(1 + \cos x) - c = 0$
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Therefore, general solution is

$$[y(1 - \cos x) - c][y(1 + \cos x) - c] = 0.$$

6. Solve: $x^2 \left(\frac{dy}{dx}\right)^2 + xy \left(\frac{dy}{dx}\right) - 6y^2 = 0$

$$x^2 p^2 + xyp - 6y^2 = 0$$

$$(xp + 3y)(xp - 2y) = 0$$

$xp + 3y = 0$ $x \frac{dy}{dx} = -3y$ $\frac{1}{y} dy = \frac{-3}{x} dx$ On integrating, $\log y = -3 \log x + \log c$ $\log y + 3 \log x = \log c$ $\log yx^3 = \log cx$ $yx^3 = c$ $x^3 y - c = 0$	$xp - 2y = 0$ $x \frac{dy}{dx} = 2y$ $\frac{1}{y} dy = \frac{2}{x} dx$ On integrating, $\log y = 2 \log x + \log c$ $\log y = \log x^2 + \log c$ $y = cx^2$ $y - cx^2 = 0$
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Therefore, the general solution is

$$(x^3 y - c)(y - cx^2) = 0$$

7. Solve: $4y^2p^2 + 2pxy(3x + 1) + 3x^3 = 0$

$$(2yp)^2 + 2yp(3x^2 + x) + 3x^3 = 0$$

$$2yp(2yp + 3x^2) + x(2yp + 3x^2) = 0$$

$$(2yp + x)(2yp + 3x^2) = 0$$

$2yp + x = 0$	$2yp + 3x^2 = 0$
$2y \frac{dy}{dx} = -x$	$2y \frac{dy}{dx} = -3x^2$
$2y dy = -x dx$	$2y dy = -3x^2 dx$
On integrating,	On integrating,
$y^2 = -\frac{x^2}{2} + c$	$y^2 = -x^3 + c$

Therefore, the general solution is

$$\left(y^2 + \frac{x^2}{2} - c\right)(x^3 + y^2 - c) = 0$$

8. Solve: $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$

$$p - \frac{1}{p} = \frac{x^2 - y^2}{xy}$$

$$\frac{p^2 - 1}{p} = \frac{x^2 - y^2}{xy}$$

$$xyp^2 - (x^2 - y^2)p - xy = 0$$

$$xp(yp - x) + y(yp - x) = 0$$

$$(xp + y)(yp - x) = 0$$

$xp + y = 0$	$yp - x = 0$
$x \frac{dy}{dx} = -y$	$y \frac{dy}{dx} = x$
$\frac{1}{y} dy = -\frac{1}{x} dx$	$y dy = x dx$
On integrating,	On integrating,
$\log y = -\log x + \log c$	$\frac{y^2}{2} = \frac{x^2}{2} + \frac{c}{2}$
$\log x + \log y = \log c$	$y^2 - x^2 = c$
$xy = c$	$y^2 - x^2 - c = 0$

Therefore, the general solution is

$$(xy - c)(y^2 - x^2 - c) = 0$$

9. Solve: $yp^2 + (x - y)p - x = 0$

$$yp(p - 1) + x(p - 1) = 0$$

$$(yp + x)(p - 1) = 0$$

$yp + x = 0$	$p - 1 = 0$
$y \frac{dy}{dx} = -x$	$\frac{dy}{dx} = 1$
$y dy = -x dx$	$dy = dx$
On integrating,	On integrating,
$\frac{y^2}{2} = -\frac{x^2}{2} + \frac{c}{2}$	$y = x + c$
$y^2 + x^2 = c$	$y - x - c = 0$
$x^2 + y^2 - c = 0$	

Therefore, the general solution is

$$(x^2 + y^2 - c)(y - x - c) = 0$$

10. Solve: $x^2p^2 + xp - (y^2 + y) = 0$

$$(x^2p^2 - y^2) + (xp - y) = 0$$

$$(xp - y)(xp + y + 1) = 0$$

$xp - y = 0$	$xp + y + 1 = 0$
$x \frac{dy}{dx} = y$	$x \frac{dy}{dx} = -y - 1$
$\frac{1}{y} dy = \frac{1}{x} dx$	$\frac{1}{y+1} dy = -\frac{1}{x} dx$
On integrating,	On integrating,
$\log y = \log x + \log c$	$\log(y + 1) = -\log x + \log c$
$\log y = \log cx$	$\log x + \log(y + 1) = \log c$
$y = cx$	$x(y + 1) = c$

Therefore, the general solution is

$$(y - cx)(xy + x - c) = 0$$

11. Solve: $xy \left(\frac{dy}{dx}\right)^2 - (x^2 + y^2) \left(\frac{dy}{dx}\right) + xy = 0$

$$xyp^2 - x^2p - y^2p + xy = 0$$

$$xp(y p - x) - y(y p - x) = 0$$

$$(y p - x)(x p - y) = 0$$

$yp - x = 0$	$xp - y = 0$
$y \frac{dy}{dx} = x$	$x \frac{dy}{dx} = y$
$y dy = x dx$	$\frac{1}{y} dy = \frac{1}{x} dx$
On integrating,	On integrating,
$\frac{y^2}{2} = \frac{x^2}{2} + \frac{c}{2}$	$\log y = \log x + \log c$
$y^2 = x^2 + c$	$\log y = \log cx$
$y^2 - x^2 - c = 0$	$y = cx$

Therefore, the general solution is

$$(y^2 - x^2 - c)(y - cx) = 0$$

12. Solve: $y \left(\frac{dy}{dx}\right)^2 + (x - y) \left(\frac{dy}{dx}\right) - x = 0$

$$yp^2 + xp - yp - x = 0$$

$$p(y p + x) - (y p + x) = 0$$

$$(p - 1)(y p + x) = 0$$

$p - 1 = 0$	$yp + x = 0$
$\frac{dy}{dx} = 1$	$y \frac{dy}{dx} = -x$
$dy = dx$	$y dy = -x dx$
On integrating,	On integrating,
$y = x + c$	$\frac{y^2}{2} = -\frac{x^2}{2} + \frac{c}{2}$
	$y^2 = -x^2 + c$

Therefore, the general solution is

$$(y - x - c)(x^2 + y^2 - c) = 0$$

13. Solve $x(y')^2 - (2x + 3y)y' + 6y = 0$

14. Solve $xp^2 + xp - yp + 1 - y = 0$

3.6 Clairaut's equation and reducible to Clairaut's equation

Introduction:

This is of the form $y = px + f(p)$. General solution is $y = cx + f(c)$.

Working rule to find singular solution:

- ❖ Differentiate general solution partially w.r.to c .
- ❖ Substitute the value of c in the general solution.

Note:

$$y = px + f(p) \text{ ---- (1)}$$

Differentiate w.r.to x ,

$$\frac{dy}{dx} = p + p'x + f'(p)p'$$

$$p'x + f'(p)p' = 0$$

$$p'[x + f'(p)] = 0$$

$$p' = 0$$

$$p = c$$

Substitute $p = c$ in (1),

$$y = cx + f(c).$$

This is the general solution.

Problems:

1. Find the general solution and the singular solution of $p = \sin(y - xp)$.

$$y - xp = \sin^{-1} p$$

$$y = xp + \sin^{-1} p$$

This is in Clairaut's form.

$$\text{General solution is } y = cx + \sin^{-1} c \text{ ----- (1)}$$

Differentiate partially w.r.to c ,

$$\begin{array}{l|l} 0 = x + \frac{1}{\sqrt{1-c^2}} & 1 - c^2 = \frac{1}{x^2} \\ \frac{1}{\sqrt{1-c^2}} = -x & c^2 = 1 - \frac{1}{x^2} \\ \sqrt{1-c^2} = -\frac{1}{x} & c = \frac{\sqrt{x^2-1}}{x} \end{array}$$

Substitute the value of c in (1).

$$y = \sqrt{x^2 - 1} + \sin^{-1} \frac{\sqrt{x^2-1}}{x}$$

This is the required singular solution.

2. Find the general solution and singular solution of $\sin px \cos y = \cos px \sin y + p$

$$\sin(px - y) = p$$

$$px - y = \sin^{-1} p$$

$$y = px - \sin^{-1} p$$

This is in Clairaut's form.

General solution is $y = cx - \sin^{-1} c$ ----- (1)

Differentiate partially w.r.to c,

$$0 = x - \frac{1}{\sqrt{1-c^2}}$$

$$\frac{1}{\sqrt{1-c^2}} = x$$

$$\sqrt{1-c^2} = \frac{1}{x}$$

$$1 - c^2 = \frac{1}{x^2}$$

$$c^2 = 1 - \frac{1}{x^2}$$

$$c = \frac{\sqrt{x^2-1}}{x}$$

Substitute the value of c in (1).

$$y = \sqrt{x^2-1} - \sin^{-1} \frac{\sqrt{x^2-1}}{x}$$

This is the required singular solution.

3. Find the general solution and the singular solution of $p = \log (px - y)$.

$$p = \log(px - y)$$

$$e^p = xp - y$$

$$y = xp - e^p$$

This is in Clairaut's form.

General solution is $y = cx - e^c$ ----- (1)

Differentiate partially w.r.to c,

$$0 = x - e^c$$

$$x = e^c$$

$$c = \log x$$

Substitute the value of c in (1).

$$y = x \log x - x$$

This is the required singular solution.

4. Find the general solution and the singular solution of $(y - px)(p - 1) = p$.

$$y - xp = \frac{p}{p-1}$$

$$y = xp + \frac{p}{p-1}$$

This is in Clairaut's form.

$$\text{General solution is } y = cx + \frac{c}{c-1} \text{ ----- (1)}$$

Differentiate partially w.r.to c,

$$0 = x + \frac{(c-1)1-c(1)}{(c-1)^2}$$

$$x = \frac{1}{(c-1)^2}$$

$$(c-1)^2 = \frac{1}{x}$$

$$c-1 = \frac{1}{\sqrt{x}}$$

$$c = 1 + \frac{1}{\sqrt{x}}$$

Substitute the value of c in (1).

$$y = x + 2\sqrt{x} + 1$$

$$y = (\sqrt{x} + 1)^2$$

This is the required singular solution.

5. Find the general solution and the singular solution of $xp^2 - yp + a = 0$.

$$yp = xp^2 + a$$

$$y = xp + \frac{a}{p}$$

This is in Clairaut's form.

$$\text{General solution is } y = cx + \frac{a}{c} \text{ ----- (1)}$$

Differentiate partially w.r.to c,

$$0 = x - \frac{a}{c^2}$$

$$x = \frac{a}{c^2}$$

$$c = \sqrt{\frac{a}{x}}$$

Substitute the value of c in (1).

$$y = 2\sqrt{ax}$$

This is the required singular solution.

6. Find the general solution and the singular solution of $xp^3 - yp^2 + 1 = 0$.

$$yp^2 = xp^3 + 1$$

$$y = xp + \frac{1}{p^2}$$

This is in Clairaut's form.

$$\text{General solution is } y = cx + \frac{1}{c^2} \text{ ----- (1)}$$

Differentiate partially w.r.to c,

$$0 = x - \frac{2}{c^3}$$

$$x = \frac{2}{c^3}$$

$$c = \left(\frac{2}{x}\right)^{\frac{1}{3}}$$

Substitute the value of c in (1).

$$y = x \left(\frac{2}{x}\right)^{\frac{1}{3}} + \left(\frac{x}{2}\right)^{\frac{2}{3}}$$

This is the required singular solution.

7. Find the general solution and the singular solution of $y + 2 \left(\frac{dy}{dx}\right)^2 = (x + 1) \frac{dy}{dx}$.

$$y = -2p^2 + (x + 1)p$$

$$y = px + (p - 2p^2)$$

This is in Clairaut's form.

$$\text{General solution is } y = cx + (c - 2c^2) \text{ ----- (1)}$$

Differentiate partially w.r.to c,

$$0 = x + 1 - 4c$$

$$c = \frac{x+1}{4}$$

Substitute the value of c in (1).

$$y = x \left(\frac{x+1}{4}\right) + \left(\frac{x+1}{4} - \frac{(x+1)^2}{8}\right)$$

$$8y = 2x(x + 1) + 2(x + 1) - (x + 1)^2$$

$$8y = (x + 1)(2x + 2 - x - 1)$$

$$8y = (x + 1)^2$$

This is the required singular solution.

8. Solve $y^2(y - xp) = x^4p^2$ using substitutions $X = \frac{1}{x}$ and $Y = \frac{1}{y}$.

$$P = \frac{dY}{dX} = \frac{-\frac{1}{y^2}dy}{-\frac{1}{x^2}dx} = \frac{y^2}{x^2}p$$

Put $x = \frac{1}{X}$, $y = \frac{1}{Y}$, $p = \frac{X^2}{Y^2}P$ in the given equation $y^2(y - xp) = x^4p^2$

$$\frac{1}{Y^2} \left(\frac{1}{Y} - \frac{1}{X} \frac{X^2}{Y^2} P \right) = \frac{1}{X^4} \left(\frac{X^2}{Y^2} P \right)^2$$

$$\frac{1}{Y^4} (Y - XP) = \frac{P^2}{Y^4}$$

$$Y - XP = P^2$$

$$Y = XP + P^2$$

This is in Clairaut's form.

General solution is

$$Y = cX + c^2 \Rightarrow \frac{1}{y} = \frac{c}{x} + c^2$$

9. Solve $(px - y)(py + x) = 2p$ by reducing into Clairaut's form, taking substitutions $X = x^2$ and $Y = y^2$.

$$P = \frac{dY}{dX} = \frac{2y dy}{2x dx} = \frac{\sqrt{Y}}{\sqrt{X}}p$$

Put $x = \sqrt{X}$, $y = \sqrt{Y}$, $p = \frac{\sqrt{X}}{\sqrt{Y}}P$ in the given equation $(px - y)(py + x) = 2p$

$$\left(\frac{\sqrt{X}}{\sqrt{Y}} P \sqrt{X} - \sqrt{Y} \right) \left(\frac{\sqrt{X}}{\sqrt{Y}} P \sqrt{Y} + \sqrt{X} \right) = 2 \frac{\sqrt{X}}{\sqrt{Y}} P$$

$$(PX - Y)(\sqrt{X}P + \sqrt{X}) = 2\sqrt{X}P$$

$$(PX - Y)(P + 1) = 2P$$

$$PX - Y = \frac{2P}{P+1}$$

$$Y = PX - \frac{2P}{P+1}$$

This is in Clairaut's form.

General solution is

$$Y = cX - \frac{2c}{c+1} \Rightarrow y^2 = cx^2 - \frac{2c}{c+1}$$

10. Solve $x^2(y - px) = p^2y$ by reducing into Clairaut's form, using the substitutions $X = x^2$ and $Y = y^2$.

$$P = \frac{dY}{dX} = \frac{2y dy}{2x dx} = \frac{\sqrt{Y}}{\sqrt{X}} p$$

Put $x = \sqrt{X}$, $y = \sqrt{Y}$, $p = \frac{\sqrt{X}}{\sqrt{Y}} P$ in the given equation $x^2(y - px) = p^2y$

$$X \left(\sqrt{Y} - \frac{\sqrt{X}}{\sqrt{Y}} p \sqrt{X} \right) = \left(\frac{\sqrt{X}}{\sqrt{Y}} P \right)^2 \sqrt{Y}$$

$$Y - pX = p^2$$

$$Y = pX + p^2$$

This is in Clairaut's form.

General solution is

$$Y = cX + c^2 \Rightarrow y^2 = cx^2 + c^2$$

11. Solve $e^{4x}(p - 1) + e^{2y}p^2 = 0$ by using substitutions $X = e^{2x}$, $Y = e^{2y}$.

$$P = \frac{dY}{dX} = \frac{2 e^{2y} dy}{2 e^{2x} dx} = \frac{Y}{X} p$$

Put $e^{2x} = X$, $e^{2y} = Y$, $p = \frac{X}{Y} P$ in the given equation $e^{4x}(p - 1) + e^{2y}p^2 = 0$

$$X^2 \left(\frac{X}{Y} P - 1 \right) + Y \left(\frac{X}{Y} P \right)^2 = 0$$

$$X^2(XP - Y) + X^2P^2 = 0$$

$$XP - Y + P^2 = 0$$

$$Y = XP + P^2$$

This is in Clairaut's form.

General solution is

$$Y = cX + c^2$$

$$\Rightarrow e^{2y} = c e^{2x} + c^2$$

12. Solve $(px + y)^2 = py^2$ by using the substitutions $X = y$ and $Y = xy$.

$$P = \frac{dY}{dX} = \frac{xdy + ydx}{dy} = \frac{xp + y}{p} = x + \frac{y}{p} = \frac{Y}{X} + \frac{X}{p}$$

$$P - \frac{Y}{X} = \frac{X}{p}$$

$$p = \frac{X}{\left(P - \frac{Y}{X}\right)}$$

Put $x = \frac{Y}{y} = \frac{Y}{X}$, $y = X$, $p = \frac{X^2}{PX - Y}$ in the given equation $(px + y)^2 = py^2$

$$\left\{\left(\frac{X^2}{PX - Y}\right)\frac{Y}{X} + X\right\}^2 = \left(\frac{X^2}{PX - Y}\right)X^2$$

$$\left\{\frac{Y}{PX - Y} + 1\right\}^2 = \frac{X^2}{PX - Y}$$

$$\left\{\frac{PX}{PX - Y}\right\}^2 = \frac{X^2}{PX - Y}$$

$$P^2 = PX - Y$$

$$Y = PX - P^2$$

This is in Clairaut's form.

General solution is

$$Y = cX - c^2$$

$$\Rightarrow xy = cy - c^2$$